Geometry-Grounded Safety Event Detection: How Calibration Error Propagates to Event-Level Accuracy

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1 Overview

Camera-based safety rules are stated in metric, real-world terms: "no one within 2 m of another moving body" or "nobody inside the restricted zone for more than T seconds." A camera, however, only observes pixels. Checking such a rule requires mapping image positions to ground-plane coordinates in meters, which is done with a planar homography H estimated from at least four image-to-floor correspondences. In practice these correspondences are clicked by hand, measured roughly, or drift when the camera is bumped, so H is never exact. Deployed vision-based safety and social-distancing systems treat the homography as given, and to our knowledge no prior work quantifies how calibration error propagates to the final *event-level* decisions. Answering that question is the goal of this project.

Our pipeline is: video $\rightarrow$ person detection $\rightarrow$ multi-object tracking $\rightarrow$ homography projection to the ground plane $\rightarrow$ event logic $\rightarrow$ event-level precision and recall. **The detector and tracker are off-the-shelf, frozen components and are not contributions of this project.** The contribution is a controlled sensitivity study of the geometric stage.

Method. We define three event types: *zone entry* (a projected foot point enters a fixed polygon), *dwell* (a person remains in a zone longer than T seconds), and *proximity* (the metric distance between two tracked people falls below d). These span the two geometric regimes of interest: zone and dwell events test a *single* projected point against a fixed boundary, while proximity tests a distance between *two* independently projected points. All three are evaluated with hysteresis and temporal debouncing, so that a track oscillating across a threshold does not generate a burst of spurious events; without this, event-level false positives are dominated by flicker rather than by geometry.

Starting from a ground-truth homography H^* and ground-truth events, we perturb each of the four image correspondences by a fixed displacement of 0, 1, 3, 5, or 10 pixels in a random direction, re-estimate H with the normalized DLT algorithm, re-project all tracks, and re-run the event logic. We study the minimal four-point case first, where the estimate is exact and has no redundancy to absorb noise; the over-determined case is examined separately as a mitigation. Each noise level is repeated over many random trials (e.g. 200), and events are matched to ground truth by temporal overlap to compute precision, recall, and F1. To isolate calibration error from perception error, every experiment is run twice: once on ground-truth tracks and once on detector/tracker output. We also analyze error as a function of distance from the camera and from the decision threshold.

Hypothesis. Proximity events depend on two projected positions, so we expect them to degrade faster than zone events. The outcome is not obvious: homography error is spatially smooth, so two nearby people may receive correlated errors that partially cancel in their difference. The final output is a practical *calibration-tolerance guideline*: for each event type, the largest correspondence error at which event F1 stays above a target (e.g. 0.95). A negative result is equally reportable. The project builds directly on the course units on model fitting and RANSAC, image transformations and homogeneous coordinates, and camera calibration.

2 Related Work

Homography estimation and uncertainty. Hartley and Zisserman (1) present the normalized DLT for homography estimation and derive first-order covariance propagation from point noise to the estimated transformation. Criminisi et al. (2) use a plane-to-plane homography for metric measurement from a single image and explicitly model measurement uncertainty, but stop at the accuracy of a measured distance rather than asking how that uncertainty changes a thresholded decision — the quantity an operator actually observes.

Metric safety monitoring. Several systems combine detection, tracking, and a ground-plane transform to flag distance violations. Punn et al. (3) used a YOLOv3 detector with DeepSORT and a bird's-eye-view transform, and Yang et al. (4) built a real-time distancing and critical-density system using inverse perspective mapping. Both report detection accuracy and qualitative results, and both treat the calibration as exact. Neither measures how an imprecise homography changes which violations are reported, which is precisely the gap this project addresses. Our perception stack follows the same tracking-by-detection pattern, held fixed throughout.

3 Datasets

Our dataset is **WILDTRACK** (5), a pedestrian dataset captured by seven calibrated, synchronized cameras with person positions annotated on a ground-plane grid. We use a *single* camera view, so the task remains monocular; the remaining views and the provided calibration serve only to establish H^* and world-frame ground truth. Because world-frame positions are annotated, ground-truth zone, dwell, and proximity events can be derived automatically rather than hand-labeled.

WILDTRACK contains pedestrians only, with no marked hazard zones or vehicles. This is not a limitation for our question. Zone polygons are *defined by us* on the ground plane, which is an advantage: we can place them at controlled distances from the camera and deliberately probe boundary regions. Proximity is measured between pairs of people, which has the same geometric structure as person-to-vehicle proximity — two independently projected ground points and one metric threshold. Dwell thresholds will be set to match the annotation rate and duration of the sequence, which we will verify before fixing T.

4 Plan of Activities

Weeks 1–2: Set up the frozen detection and tracking pipeline; compute H^* and derive ground-truth events on WILDTRACK. **Weeks 3–4:** Implement the event logic with hysteresis, the perturbation harness, and the event-level evaluation; run the core sweep on ground-truth tracks. **Weeks 5–6:** Repeat on predicted tracks; distance-dependent analysis. **Weeks 7–8:** Zone-placement and reference-point-layout experiments; over-determined estimation as a mitigation. **Week 9:** Final analysis, report, and presentation.

By the final report we will deliver precision/recall-versus-ϵ curves for all three event types, under both ground-truth and predicted tracks, together with the resulting calibration-tolerance guideline.

Division of work. *Dhanya Kini Bailoor*: homography estimation, the perturbation harness, and the sensitivity analysis. *Tejaswini Sarikonda*: detection and tracking pipeline, dataset preparation, event logic, ground-truth event derivation, and the event-level evaluation metrics. Both members share error analysis, experiment design, and writing.

References

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